An Analytical Model for Reverberation Time Prediction of Indoor Scenarios

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***Abstract*—A novel analytical model is proposed for rapid and accurate prediction of the reverberation time (RT) in rooms. Through rigorously considering walls with finite thicknesses composed of multiple layers of building materials, this model successfully captures the RT fluctuations that were overlooked in previous models. Unlike previous RT models that were fitted based on data measured within specific frequency ranges, the proposed analytical model is capable of predicting RT across a wide frequency spectrum. The model is experimentally validated in two different residential building rooms over a frequency range of (1 to 6) GHz and using the measured data in a laboratory from literature within (1 to 20) GHz. The predicted results using the proposed model are in good agreement with those obtained from existing models. Importantly, the proposed model accurately predicts RT fluctuations, consistent with measured results in the literature. Compared to traditional measurement methods, this analytical model offers a more efficient alternative for predicting RT in indoor scenarios.**

***Index Terms*—Analytical model, building materials, indoor scenarios, quality factor, reverberation time.**

I. Introduction

MODELS describing the time dispersion of indoor propagation channels have been widely utilized in various applications, including radio channel sounding [1], [2], [3], [4], [5], [6], [7], path loss [8], [9], [10], and arrival rate [11] predictions, localization technologies [12], and human exposure analysis [13]. One important metric used to characterize time dispersion is the power delay profile (PDP) tail, which is also referred to as the reverberation time (RT) [14]. Experimental evidence has demonstrated that reverberation occurs in a room as a result of diffuse reflections of electromagnetic (EM) waves from walls and furniture constructed using different materials [15]. Therefore, the RT can be obtained by measuring the PDP and extracting its exponentially decaying tail.

Several previous studies have reported on the measurement of RT in different indoor scenarios. For example, one study focuses on an office room with hard paper partitions [14], while others examine various indoor environments in new and old buildings [16], as well as a large industrial hall [17]. The polarization characteristics of RT have been discussed in [17]. Furthermore, the presence of people is found to have an effect on RT, as suggested by the results in [18]. Although these studies offer effective methods for RT measurement, their primary focus is on measuring RT at a single frequency point.

Subsequently, a frequency-dependent RT model is developed for indoor scenarios within the frequency range of (2 to 10) GHz. This model is based on data fitting at eight frequency points [19]. Another study proposes a modified RT model for a frequency range of (1 to 40) GHz. This model incorporates the probability distribution of the number of reflections instead of only a constant number [20]. Both studies highlight the dependence of RT on frequency, room volume and surface area. It is important to note that these RT models require extensive measurements and data fitting, which in turn demands significant time and human resources.

In contrast, an analytical RT model offers a more efficient alternative. Previous research has presented several analytical solutions for RT [21], [22], primarily based on the theory of electrically large cavities [23], [24]. However, these analytical models make the assumption that the walls are either infinitely thick or impenetrable, which is not the case in real-world indoor environments where the thickness of a building is finite. The thickness of walls and the dielectric properties of their materials significantly impact the propagation of EM waves [25], [26], [27], [28], [29], [30]. Notably, partitions made of materials like gypsum plaster, wood, glass, and others generally have a thinner profile compared to load-bearing walls. At low frequencies, EM waves can easily penetrate these thinner walls, resulting in multiple reflections on their front and rear surfaces. Consequently, these reflections cause variations in the RT as the frequency changes, which should be taken into consideration in the RT model.

In this letter, a novel analytical RT model for indoor scenarios is presented. Enhanced accuracy in predicting RT over a wide frequency range is achieved through comprehensive modeling of building walls with finite thickness. The incorporation of a tri-layered structure of walls in the model enables the estimation of fluctuation characteristics in RT, a feature that has been overlooked in previous models.

II. Proposed Analytical Model

Based on the theory of electrically large cavities, the quality factor Q is expressed as follows [24]:

$$Q = \omega \cdot \tau \tag{1}$$

where $\omega = 2\pi f$ is the angular frequency (rad/s) and τ is RT. When the cavity is uniformly stirred, Q can be described by the relationship between average power density (S_c) and dissipated

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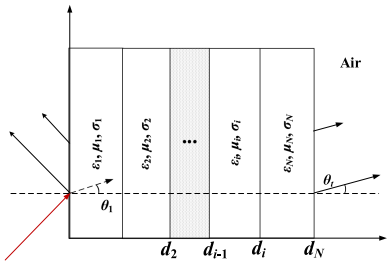


Fig. 1. Propagation of oblique incident wave through walls composed of N-layer building materials.

power (P_{d})

$$Q = \omega U / P_{\mathrm{d}} = \omega S_c V / P_{\mathrm{d}} c \tag{2}$$

where U denotes the steady state energy in the cavity, V and c represent the cavity volume (m^3) and light velocity (m/s) in free space, respectively. Four types of losses are accounted for by the dissipated power in the cavity, as depicted in [24]

$$P_{\mathrm{d}} = P_{\mathrm{d}1} + P_{\mathrm{d}2} + P_{\mathrm{d}3} + P_{\mathrm{d}4} \tag{3}$$

where $P_{\mathrm{d}\,i}$ $(i = 1, 2, 3, 4)$ sequentially denotes power dissipated in cavity walls, absorbed by objects inside the cavity, leaked through apertures and dissipated on loads of receiving antennas. Since the walls in buildings are different from those of metallic enclosures, here we focus only on the power dissipated in the walls.

The power dissipated in the walls of an electrically large cavity can be calculated as [23]

$$P_{\mathrm{d}1} = S_c \cdot \langle \sigma_{\mathrm{wall}} \rangle \tag{4}$$

where σ_{wall} denotes the absorption cross section of walls, and $\langle \cdot \rangle$ indicates an average over all incident angles and polarizations [24]. The $\langle \sigma_{\mathrm{wall}} \rangle$ can then be written as

$$\langle \sigma_{\mathrm{wall}} \rangle = \frac{1}{2} S \int_0^{\frac{\pi}{2}} \left[1 - \frac{1}{2} \left(|\Gamma_\perp|^2 + |\Gamma_\parallel|^2 \right) \cos\theta \sin\theta d\theta \right] \tag{5}$$

where S is the total surface area of the cavity walls (m^2), $\Gamma_\perp$ and $\Gamma_\parallel$ represent the reflection coefficient for perpendicular and parallel polarizations.

EM waves cannot penetrate through a metallic cavity wall or a highly lossy wall. Therefore, the reflection coefficients in (5) can be determined using the reflection coefficient (Γ_0) at the interface between two semi-infinitely large media, as depicted in previous research [21]. However, in indoor environments such as residential buildings, load-bearing walls are typically constructed using concrete and brick, while partition walls are made of materials like glass, plasterboard, wood, etc. Moreover, these walls have limited thickness. Thus, the Sub -6 GHz EM waves can easily penetrate through these walls. Consequently, it is not reasonable to approximate the total reflection coefficients using Γ_0.

In the case of a wall with finite thickness, the reflected EM waves at the interface include not only the initial reflection at the front interface but also multiple reflections caused by the front and rear surfaces. This issue becomes more pronounced when the wall is made up of multiple layers of materials, as depicted in Fig. 1.

Consequently, it is crucial to consider both powers dissipated due to the penetration of walls (σ_{wall}) and transmission through the walls (σ_{tran}) when calculating $P_{\mathrm{d}1}$

$$P_{\mathrm{d}1}(f, d) = S_c \cdot (\langle \sigma_{\mathrm{wall}} \rangle + \langle \sigma_{\mathrm{tran}} \rangle) \tag{6}$$

where σ_{wall} and σ_{tran} represent the absorption and transmission cross sections of walls and can be calculated using reflection (Γ) and transmission (T) coefficients for multiple layers.

$$\begin{cases} \langle \sigma_{\mathrm{wall}} \rangle = \frac{1}{2} S \int_0^{\frac{\pi}{2}} \left[1 - \frac{1}{2} \left(|\Gamma_\perp|^2 + |\Gamma_\parallel|^2 \right. \right. \\ \qquad \left. \left. + |\mathrm{T}_\perp|^2 + |\mathrm{T}_\parallel|^2 \right) \cos\theta \sin\theta d\theta \right] \\ \langle \sigma_{\mathrm{tran}} \rangle = \frac{1}{2} S \int_0^{\frac{\pi}{2}} \left[\frac{1}{2} \left(|\mathrm{T}_\perp|^2 + |\mathrm{T}_\parallel|^2 \right) \cos\theta \sin\theta d\theta \right] \end{cases} \tag{7}$$

where $\Gamma_\perp, \mathrm{T}_\perp, \Gamma_\parallel, \mathrm{T}_\parallel$ are total reflection and transmission coefficients for perpendicular and parallel polarizations, respectively, and can be calculated as follows [31]:

$$\Gamma = \frac{M_0^{k=1}}{M_0^{k=2}}, \mathrm{T} = \frac{1}{M_0^{k=2}}. \tag{8}$$

The parameters ${M_0}^k$ is determined using the recursive formula

$$M_j^k = \frac{e^{(-1)^k \psi_j}}{2} \left[M_{j+1}^{k=1} \left(1 - (-1)^k N_{j+1} \right) + M_{j+1}^{k=2} \left(1 + (-1)^k N_{j+1} \right) \right] \tag{9}$$

where

$$M_{N+1}^{k=1} = 0, \ M_{N+1}^{k=2} = 1, \psi_j = d_j \gamma_j \cos\theta_j \tag{10}$$

and the parameter N_{j+1} for perpendicular ($\perp$) and parallel ($\parallel$) polarizations is given as

$$N_{j+1} = \begin{cases} \frac{\cos\theta_{j+1}}{\cos\theta_j} \cdot \frac{\eta_j}{\eta_{j+1}} \ (\perp) \\ \frac{\cos\theta_{j+1}}{\cos\theta_j} \cdot \frac{\eta_{j+1}}{\eta_j} \ (\parallel) \end{cases} \tag{11}$$

where θ_j and η_j are complex angles of refraction and intrinsic impedance in the jth layer, respectively.

Notably, the magnitude of the reflection coefficient magnitude may oscillate when EM waves impinge on a low lossy wall with finite thickness [25]. This is caused by the superposition and cancellation of EM waves reflected from the front and rear surfaces of the wall at different frequencies. These oscillations can lead to frequency-dependent fluctuations of RT.

Substituting (3)–(11) into (2) leads to

$$Q(f, d) = \frac{2\pi V}{\lambda \left(\langle \sigma_{\mathrm{wall}} \rangle + \langle \sigma_{\mathrm{tran}} \rangle + \langle \sigma_3 \rangle + \langle \sigma_4 \rangle \right)} \tag{12}$$

where λ represents the wavelength, $\langle \sigma_3 \rangle$ denotes the dissipation cross section of windows and doors, and $\langle \sigma_4 \rangle$ refers to the average effective area of antennas. The estimation of $\langle \sigma_3 \rangle$ and $\langle \sigma_4 \rangle$ can be achieved using the method outlined in [24].

Based on these, the Q-factor density (Q_{d}), which is defined as Q per volume per area in [19] can be given as

$$Q_{\mathrm{d}}(f, d) = \frac{2\pi S}{\lambda \cdot \left(\langle \sigma_{\mathrm{wall}} \rangle + \langle \sigma_{\mathrm{tran}} \rangle + \langle \sigma_3 \rangle + \langle \sigma_4 \rangle \right)}. \tag{13}$$

Substituting (13) into (1) leads to

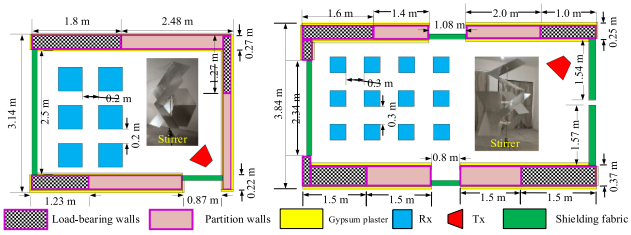
$$\tau(f, d) = \frac{V}{c \cdot \left(\langle \sigma_{\mathrm{wall}} \rangle + \langle \sigma_{\mathrm{tran}} \rangle + \langle \sigma_3 \rangle + \langle \sigma_4 \rangle \right)}. \tag{14}$$

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(a) (b)

Fig. 2. Measurement scenarios of (a) room A and (b) room B.



(a) (b)

Fig. 3. Measurement setup scheme for (a) room A and (b) room B.

Thus, the analytical solution for RT in rooms can be obtained by evaluating $\langle \sigma_{\text{wall}} \rangle$ and $\langle \sigma_{\text{tran}} \rangle$ through the calculation of reflection (Γ) and transmission (T) for all walls, ceiling and floor, and $\langle \sigma_3 \rangle$, as well as $\langle \sigma_4 \rangle$. As a forementioned, Γ and T are functions of frequency and wall parameters, including building materials type, number of layers, and thickness.

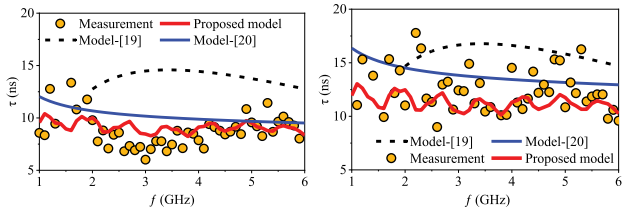
III. Experimental Scenario and Setup

A. Scenario

To validate the proposed analytical model, frequency-dependent RT measurements were conducted in two bare residential rooms with different dimensions. Room A measures approximately 4.28 m $\times$ 3.14 m $\times$ 2.78 m, resulting in a volume of 37.36 m^3 and a total surface area of 63.79 m^2. Room B has dimensions of 6.80 m $\times$ 3.85 m $\times$ 2.78 m, resulting in a volume of 72.78 m^3 and a total surface area of 88.70 m^2. Both rooms feature concrete floors and ceilings, showcasing the details of the load-bearing and partition walls in Fig. 3. The walls, floors and ceilings are regarded as tri-layer structures in the following calculation, with a concrete/plasterboard layer sandwiched between two 5 mm-thick gypsum plaster layers.

B. Experimental Setup

The RT is typically obtained by measuring the average transmission coefficient between a stationary transmitting antenna (Tx) and a receiving antenna (Rx) positioned uniformly throughout the room [32], [33], [34]. In this study, the RT is determined by measuring the Q in the frequency domain following the approach outlined in [35]. To replicate the reverberation effect in an empty room, a mode stirrer is employed in combination with a position-stirring approach. This method involves changing Rx position, as depicted in Figs. 2 and 3.



(a) (b)

Fig. 4. RT in (a) room A and (b) room B.

The mode stirrer shown in Fig. 2 is positioned approximately 0.5 m away from the walls and has a height of 1.8 m with a rotational diameter of 1.2 m. It consists of six lightweight polystyrene boards covered with aluminum foil, each measuring 2 mm in thickness and 90 cm $\times$ 60 cm in dimension.

Since the aim of measurement is to validate the proposed model, the doors and windows of the rooms are covered with shielding fabrics to ensure that EM waves can only enter or exit the room through the walls, ceiling and floor. This meticulous setup allows for a thorough examination of the impact of wall loss on both Q and RT.

C. Measurement

A pair of wideband double-ridged horn antennas are employed to measure the transmission coefficient (S_{21}) in the frequency range from 1 GHz to 6 GHz with a step of 100 MHz. The Tx is connected to a Rohde & Schwarz signal generator (R&S SGS100A 1 MHz to 12.75 GHz), while the Rx is connected to a spectrum analyzer (R&S FSH13 9 kHz to 13 GHz), as illustrated in Fig. 2. The stirrer rotates at a consistent speed of 2.77 degrees per second. At each frequency point, the $|S_{21}|$ is measured at 260 different stirrer positions. Consequently, a total of 13 260 samples are recorded for each Rx position. For further details regarding the complete measurement configuration, please refer to [36].

As shown in Fig. 3, the room floors are divided into multiple 1 $\times$ 1 m^2 grids in the measurement. The Tx is positioned at a fixed height of 1.5 m outside the measurement area, while the Rx is placed within the grids at heights of 1.5 m and 1.8 m, respectively.

IV. Results and Validation of the Model

A. Validation With Room A and Room B

First, the proposed RT and Q_{d} model is validated by conducting experiments in rooms A and B. The permittivity of the materials in the tri-layer walls is determined using the Cole-Cole model presented in [37], which investigates the permittivity of 20 common building materials in the frequency range of (0.2 to 67) GHz. The measured and predicted results of RT and Q_{d} are compared to the results calculated using the models from [19] and [20]. For brevity, these models are referred to as model-[19] and model-[20], as shown in Figs. 4 and 5.

In Fig. 4, the predicted RTs obtained from the proposed analytical model align well with the results calculated using model [20] for both rooms. Moreover, the measured RTs also align closely with the predicted results from the analytical model and model-[20], further validating the proposed analytical

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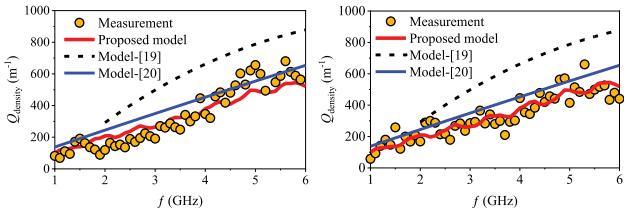


Fig. 5. Q_{d} of (a) room A and (b) room B.

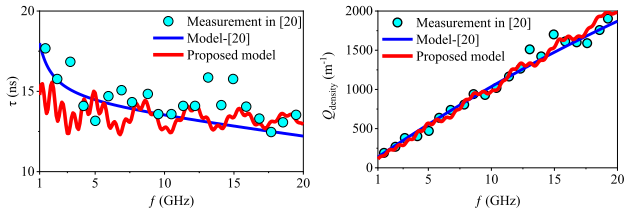


Fig. 6. (a) RT and (b) Q_{d} of the laboratory in [20].

TABLE I
RMSEs OF MODEL-[20] AND PROPOSED MODEL FOR ROOMS A AND B

Prediction method	RT - RMSE		Q_d - RMSE	
	Room A	Room B	Room A	Room B
Model-[20]	2.08	2.76	100.44	91.79
Proposed model	1.47	2.73	88.25	58.14

model. In contrast, the results obtained from model-[19] are slightly larger, although they exhibit the same variation trend with frequency. Notably, the measured RT in room B fluctuates more significantly compared to room A, which can be attributed to the more extensive use of shielding fabrics in room B, as depicted in Fig. 3.

Fig. 5 presents the measured Q_{d} values and their predictions using the three models within the frequency range of 1–6 GHz. It is evitable that all models display an increasing trend with frequency. A closer examination reveals good agreement between the results predicted by the analytical model and the measured data, as well as the results obtained from model-[20], which further validates the accuracy of the analytical model. Again, the model-[19] predicts larger Q_{d} values compared to both the analytical model and model-[20].

Fig. 5 does not reveal a noticeable difference in Q_{d} between the two rooms. However, Fig. 4 shows that the RT in room B is larger than in room A, which can be attributed to the fact that the ratio of volume over surface area (V/S) is greater in room B. This observation exactly agrees with the relationship established in [19] between RT, Q_{d} and the V/S ratio

$$\tau = \frac{Q_{\mathrm{d}}}{2\pi f} \times \frac{V}{S}. \tag{15}$$

To quantitatively evaluate the accuracy of the proposed analytical model, the root mean square errors (RMSEs) of predicted RTs and Q_{d}s compared to the measurement are calculated using a total of 51 frequency points across the frequency band, and results are listed in Table I. The smaller RMSEs validate the accuracy of the proposed model.

B. Validation With the Scenarios in Literature

To further validate the proposed model, both RT and Q_{d} of the laboratory described in [20] are calculated in the frequency range of (1 to 20) GHz. The laboratory has a volume of 160 m^3 and a surface area of 195 m^2. The walls are constructed with plasterboard, while the floor and ceiling are made of concrete. Since the thickness is not specified in [20], a typical tri-layered structure with a thickness of 250 mm is used in the calculations.

TABLE II
RMSEs OF MODEL-[20] AND PROPOSED MODEL FOR THE ROOM IN LITERATURE

Prediction Model	RT - RMSE	Q_d - RMSE
	Laboratory in [20]	Laboratory in [20]
Model-[20]	1.23	82.15
Proposed model	1.79	92.88

The results are then compared to the measured and predicted data presented in [20], as illustrated in Fig. 6(a) and (b). The calculated Q_{d} using the analytical model agrees very well with the measured and predicted results in [20], as depicted in Fig. 6(b). Regarding the RT shown in Fig. 6(a), the results from the proposed model are in good accordance with the measured data, particularly in the fluctuation. This can be attributed to the fact that the wall material used in the rooms is plasterboard, which allows penetration of EM waves, resulting in oscillation in reflection, as discussed above. However, this phenomenon is overlooked by model-[20], which is a fitting formula based on measurement.

Table II summarizes the RMSEs obtained by comparing the predicted results to the measurement in [20] at 191 frequency points. The RMSE of the proposed model is close to the one of model-[20]. The slight discrepancies between them primarily can be attributed to the fact that model-[20] is fitted based on the measured data, and the actual wall material properties and thicknesses differ from those used in the analytical model. Notably, this analytical model is remarkably efficient in RT prediction compared to the experimental approach.

V. CONCLUSION

A novel analytical model is presented for the rapid and accurate prediction of RT in indoor scenarios. The model takes into account the finite-thick walls, ceiling and floor, which are composed of multiple layers of building materials. The accuracy of the proposed model is quantitively validated through experimental results, as well as comparisons with fitted models and measured data from literatures. It is worth noting that the proposed model can predict the RT fluctuations, a feature previously overlooked in existing RT models. This model is applicable to indoor environments where reverberation occurs over a wide frequency range. Despite the presence of furniture and other objects, the proposed model remains valid by incorporating the dissipation effects of those objects.

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